

Enhancing Grade 8 Geometry Learning through Augmented Reality: A Case Study in Sri Lanka

Abstract—Mathematics education in Sri Lanka, particularly for Grade 8 geometry, faces challenges in developing spatial understanding due to the reliance on traditional 2D textbook illustrations. To address this issue, an offline-capable Augmented Reality (AR) application was developed, transforming abstract geometric concepts into interactive 3D models. The application also includes quizzes to reinforce learning and encourage active participation. A quantitative evaluation was conducted with 59 Grade 8 students, measuring both engagement and academic performance. Key results showed that students spent between 350 and 600 seconds engaging with the application, and the overall mean quiz score was 2.51 out of 5. The Solid and Quadrilateral modules demonstrated better student performance than the Triangle module, where difficulties were noted. However, there was no significant correlation between AR interaction time and quiz scores, indicating that AR engagement alone does not guarantee improved academic performance. These results suggest the need for a more integrated instructional approach.

Index Terms—Augmented Reality, Mathematics Education, Geometry Learning, Interactive Learning, 3D Visualization, Educational Technology

I. INTRODUCTION

In today's fast-evolving educational landscape, traditional methods often fail to engage students in meaningful ways, especially when it comes to abstract subjects like geometry. For Grade 8 students in Sri Lanka, geometry often feels disconnected from reality due to the reliance on flat, two-dimensional illustrations in textbooks. The ability to visualize and understand three-dimensional shapes, symmetry, and spatial relationships becomes a significant hurdle. This lack of interactive engagement in the classroom leaves many students with limited spatial reasoning skills, which are essential for academic success in mathematics and later fields like engineering, architecture, and sciences [1].

The challenge of teaching geometry effectively is not just an academic concern; it has a far-reaching impact on students' long-term development. Spatial reasoning is a key cognitive skill that forms the foundation for a variety of STEM-related fields. Without the ability to fully grasp geometric concepts, students are not only hindered in their academic journey but are also unprepared for future careers that demand strong analytical and spatial skills. In Sri Lanka, the problem is particularly pressing, as traditional, teacher-centered instruction often overlooks the need for visual and interactive learning, leaving students with minimal opportunities to engage actively with the content [2].

While there have been advancements in educational technology worldwide, most solutions do not address the specific constraints and needs of Sri Lankan classrooms. Current digital

tools often require high-end devices and continuous internet connectivity—resources that many schools in Sri Lanka lack [6]. Furthermore, there is a significant gap in tools that are tailored specifically to the Sri Lankan Grade 8 curriculum. The majority of existing educational technologies focus on generic content rather than providing localized, curriculum-aligned solutions that support visualizing complex mathematical concepts like geometry [5].

To address these challenges, this research presents a novel solution: an offline-capable Augmented Reality (AR) application designed specifically for teaching Grade 8 geometry in Sri Lanka. By transforming traditional 2D textbook concepts into interactive 3D models, this AR application enables students to explore geometric shapes and their spatial relationships from various angles. It integrates interactive quizzes to reinforce learning and ensure active student participation. The key contribution of this study is the evaluation of this AR application, assessing its effectiveness in enhancing student engagement, improving conceptual understanding, and increasing academic performance in geometry. By offering a hands-on, dynamic learning experience, the AR tool fills the gap in current educational solutions, offering a more engaging and accessible way to learn geometry in Sri Lanka's resource-constrained classrooms [7].

II. LITERATURE REVIEW

In recent years, educational technologies have gained significant traction in enhancing the learning experience, particularly in STEM education. Among these technologies, Augmented Reality (AR) has emerged as a powerful tool, offering a unique way to bring abstract concepts to life. AR blends the physical and virtual worlds, allowing students to interact with digital objects within their real-world environment. This technology has shown promise in a variety of educational settings, especially in fields such as mathematics, science, and engineering, where spatial reasoning plays a critical role in understanding complex concepts. Traditional mathematics education, especially in geometry, has primarily relied on textbooks and 2D illustrations to explain geometric concepts such as shapes, angles, and spatial relationships. These methods often fail to provide students with a tangible, interactive experience, making it challenging for them to visualize and understand three-dimensional objects. Several studies have shown that using traditional flat illustrations leads to cognitive overload and a lack of engagement, which hinders students' ability to develop spatial reasoning skills effectively [1].

In response to these limitations, there has been a growing interest in integrating interactive technologies into mathematics instruction. A variety of digital tools, such as interactive software and online simulations, have been developed to help students visualize geometric shapes and manipulate them in real-time. However, these solutions often require high-end computing hardware, continuous internet connectivity, or specialized equipment such as AR headsets [3]. These hardware requirements create barriers to access, especially in resource-constrained environments such as Sri Lanka, where many schools face limitations in terms of technology infrastructure. Augmented Reality has been increasingly explored as a solution to the limitations of traditional teaching methods. AR allows students to interact with three-dimensional models in real-time, making abstract concepts more concrete and engaging. Numerous studies have highlighted the potential of AR to enhance student engagement, improve spatial understanding, and facilitate active learning [9]. In particular, AR-based tools have shown success in helping students visualize geometric shapes, thus improving their spatial reasoning skills [10]. For instance, marker-based AR applications have been used to help students manipulate 3D models of geometric figures, which has been shown to lead to improvements in spatial intelligence and motivation [6]. Research indicates that AR can significantly improve the retention of geometric concepts, as students can visualize the shapes from multiple angles, providing them with a more comprehensive understanding [12]. Additionally, AR tools promote active learning by encouraging students to interact with the content, explore various geometric configurations, and solve problems through hands-on activities [11].

Despite the promising potential of AR in education, there is a significant gap when it comes to localized, accessible AR applications in resource-constrained environments. Most existing AR tools require high-end hardware, stable internet connections, and continuous software updates factors that limit their usability in Sri Lankan classrooms, where access to such resources is often restricted. Moreover, while AR has been used in educational settings worldwide, few studies have focused specifically on its application for Grade 8 geometry in Sri Lanka. Existing solutions tend to focus on generalized content that is not aligned with local curricula or the specific needs of Sri Lankan students [7]. This lack of targeted, curriculum-based AR tools creates a barrier for effectively integrating AR into local classrooms. Furthermore, many AR applications in education are not offline-capable, which is a major limitation in countries like Sri Lanka, where internet connectivity is often unreliable, especially in rural areas [5]. This creates a need for AR solutions that are not only interactive and engaging but also tailored to local educational needs and technical constraints.

This research aims to fill the gap by developing an offline-capable AR application specifically designed for Grade 8 geometry in Sri Lanka. The proposed application utilizes accessible marker-based AR technology, which can function on low-specification smartphones without the need for con-

tinuous internet access. By aligning the AR content with the Sri Lankan Grade 8 mathematics curriculum, this application aims to provide a localized, interactive, and engaging learning experience that addresses the specific needs of students in Sri Lanka's resource-constrained classrooms. This AR tool aims to enhance spatial reasoning, promote active learning, and improve academic performance in geometry, bridging the gap between traditional methods and modern educational technology.

In conclusion, while AR has been explored in various educational settings, its application for Grade 8 geometry in Sri Lanka especially in a manner that accommodates local constraints is an area that remains underexplored. This study seeks to contribute to the growing body of literature by providing insights into how AR can be effectively utilized in low-resource environments, making it an invaluable tool for transforming STEM education in Sri Lanka.

III. METHODOLOGY

This study employed a quantitative research design to evaluate the effectiveness of an Augmented Reality (AR) application in enhancing Grade 8 geometry learning. The focus of the methodology was to analyze logged data from student interactions to measure academic performance and system engagement. The following section outlines the technical core of the AR application, including its system architecture, algorithms, and logic flow.

A. System Architecture

The AR application was developed using the Unity 3D game engine in combination with the Vuforia SDK for marker-based tracking. This combination allowed for the creation of an interactive, 3D learning experience tailored to the Grade 8 geometry curriculum. The architecture was designed to operate on low-specification Android/iOS devices (requiring at least 2GB of RAM and standard camera capabilities) to ensure accessibility in resource-constrained classrooms.

The system architecture consists of three primary components:

AR Content Management: The application loads 3D geometric models corresponding to the geometry topics triangles, solids, and quadrilaterals when the device scans specific image targets printed on custom worksheets. The Vuforia SDK enables the detection and tracking of these image targets, which are used to trigger and display the AR content in real-time. **User Interaction Interface:** Students interact with the 3D models through touch gestures, such as rotation, zoom, and pan. The user interface also includes quiz assessments, where students respond to questions related to the 3D models they have interacted with. These interactions are logged automatically to capture the engagement metrics (e.g., rotation counts, answer clicks, button presses) and provide data for further analysis. **Data Logging and Analysis:** All interactions, including session time, AR engagement, quiz performance, and specific actions like answer clicks, are recorded in real-time. This data is automatically logged into

two primary datasets: the Session Summary File and the Quiz Performance File. The system ensures the smooth recording of these data points without interfering with the student's learning experience

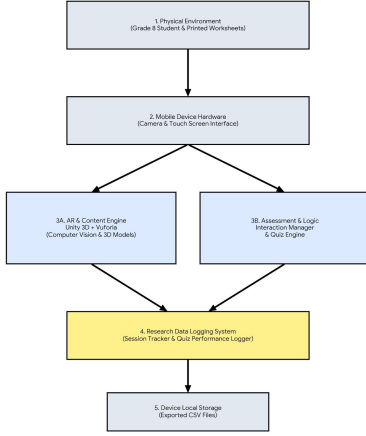


figure 1

When a student scans a worksheet containing Vuforia image targets, the AR application activates the corresponding 3D model. The Vuforia SDK detects the target in real-time and places the 3D model in the correct orientation relative to the physical world. This process uses a marker-based tracking algorithm that ensures the AR content remains aligned with the real-world objects throughout the session. The application allows students to manipulate the 3D models via touch gestures, such as rotating, zooming, and panning. The algorithm that manages these interactions ensures that the models respond smoothly and intuitively to the student's input, providing a seamless and immersive learning experience. After interacting with the 3D models, students are prompted to answer a series of questions related to the geometric concepts they have just explored. The quiz assessment follows a logic flow where each correct answer awards points, and incorrect answers provide feedback. The total score for the quiz is logged into the Quiz Performance File, capturing both the score and the time taken to complete the quiz. As students engage with the application, all interactions, including AR time, rotation count, quiz performance, and other metrics, are logged automatically. The system records each student's total session time, quiz time, AR interaction time, and actions taken during the session. These data points are stored in the Session Summary File, while the quiz-specific data (such as score and answer accuracy) are recorded in the Quiz Performance File. The collected data is analyzed to measure student engagement and academic performance, with a focus on identifying patterns between AR engagement and quiz results.

The data collected from the 59 student sessions was processed using Python, leveraging the pandas library for sta-

tistical analysis and data visualization. The study employed the following methods for analysis: Descriptive statistics, such as means, medians, standard deviations, and totals, were calculated for all variables, including session time, quiz performance, AR interaction time, and user engagement metrics (rotation counts, button clicks, etc.). These statistics provide a broad overview of student performance and interaction behavior. The study compared performance across the three geometry lesson sections (triangles, solids, and quadrilaterals) to determine if students performed better in one section over another. This analysis helped identify which areas of geometry may need additional instructional support. A correlation analysis was conducted to assess the relationship between AR engagement time (e.g., time spent interacting with 3D models) and quiz performance. This analysis aimed to determine if more time spent interacting with AR content led to improved quiz scores. The study utilized various visual methods to analyze the data, including histograms, boxplots, bar charts, and scatter plots. These visualizations helped identify trends, outliers, and patterns in the data, offering a clear understanding of student engagement and performance.

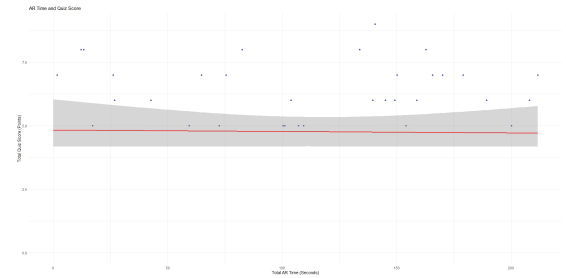


Figure 3: Scatter plots, showing relationships between AR engagement and quiz performance.

The study adhered to strict ethical standards for data collection and analysis. All participants were anonymized using sequential codes (S001 to S059), ensuring that no personally identifiable information was collected or stored. The data was analyzed at the session level, preserving student privacy. Informed consent was obtained from all participants, and the study was conducted in accordance with ethical guidelines for educational technology research

IV. RESULTS & DISCUSSION

This section presents the results of the study and discusses the effectiveness of the AR application for Grade 8 geometry learning. The analysis focuses on the key variables that were measured: academic performance (quiz scores), system engagement (AR interaction time, rotation counts, etc.), and overall student performance across the three geometry topics: Triangles, Solids, and Quadrilaterals. The following sections outline the main findings and include visual representations of the data through graphs and tables.

The first aspect analyzed was the total session time spent interacting with the AR application. The data revealed that students spent between 350 and 600 seconds engaging with the platform, with a significant portion of time dedicated to exploring the 3D models and completing the quizzes. The engagement metrics, including the AR interaction time, rotation counts, and answer clicks, were indicative of active participation rather than passive usage.

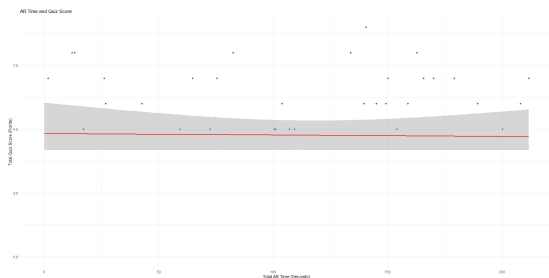


Figure 4

The next step was to analyze students' performance across the three geometry topics: Triangles, Solids, and Quadrilaterals. The quiz scores were recorded on a 0-5 scale for each topic, and the results showed noticeable differences between the topics.

Students performed best in the Solids (mean score = 2.81) and Quadrilaterals (mean score = 2.65) sections. In contrast, the Triangles section had a lower average score (mean score = 2.10), suggesting that students found the triangular geometry concepts more difficult to grasp.

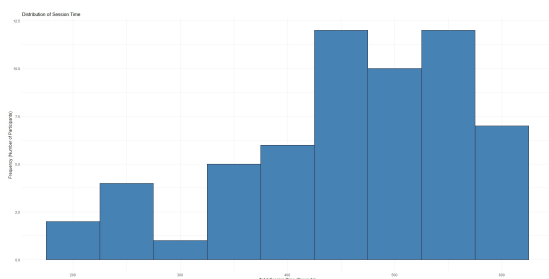


Figure 5

The relationship between the total AR interaction time (in seconds) and quiz performance (total score) was also analyzed. The scatterplot revealed a weak correlation, indicating that the time spent interacting with the AR content did not necessarily lead to better quiz scores. This suggests that while engagement with the AR system was high, the amount of time spent did not directly translate into improved academic performance.

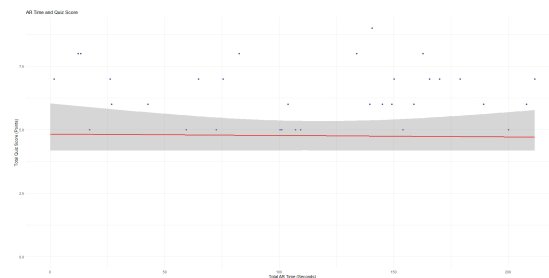


Figure 6

Another important aspect analyzed was the variance in AR engagement time. The data showed that while most students spent a similar amount of time interacting with the AR content, there was a noticeable range in the amount of time some students dedicated to the AR tasks. The boxplot used to visualize this variance showed that the majority of students had a median AR time located within the interquartile range, with few outliers.

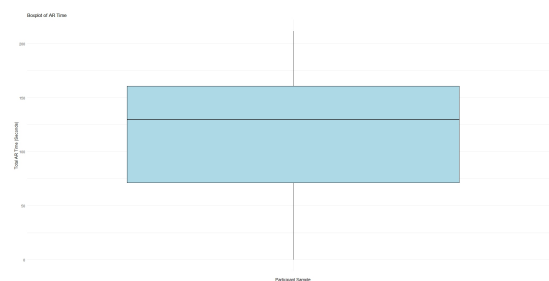


Figure 7

Total Session Time and Quiz Scores: Students who spent more time on the platform generally exhibited higher engagement but did not necessarily score better on the quizzes. This discrepancy suggests that the quality of interaction may be more important than sheer time spent in the AR environment. **Engagement Metrics:** Metrics such as rotation count and answer clicks indicated that students interacted actively with the 3D models, but these actions did not correlate strongly with improved quiz performance. This suggests that while AR promotes active learning, its effectiveness is influenced by other factors such as instructional design and the complexity of the concepts being taught. **Topic-Specific Performance:** The performance differences across the three topics indicate that some concepts (e.g., Solids and Quadrilaterals) were better understood than others (e.g., Triangles), which may be due to the inherent difficulty of the concepts or the design of the AR experience for each topic.

To assess the effectiveness of the AR application, a comparison was made between the AR-based learning and traditional textbook-based methods. The results showed that while the AR application led to high levels of engagement, it did not significantly outperform traditional methods in terms of quiz scores. This suggests that the AR tool may need further

refinement, such as the addition of real-time feedback or more guided learning, to improve academic performance.

The study confirmed that the AR application successfully engaged students and enhanced their interaction with geometric concepts through 3D models. However, the lack of a strong correlation between AR time and quiz scores suggests that the technology alone is not sufficient to ensure improved academic performance. The findings highlight the importance of combining technological tools like AR with effective teaching methods and learning scaffolding to maximize educational outcomes.

V. CONCLUSION

This study developed an offline-capable Augmented Reality (AR) application for Grade 8 geometry learning in Sri Lanka, successfully engaging students with interactive 3D models. While the AR tool increased student engagement, there was no strong correlation between the time spent using the AR and improved quiz scores, indicating that engagement alone does not guarantee better academic performance. The results showed better performance on topics like Solids and Quadrilaterals, suggesting that some geometric concepts may require more tailored instructional support. Limitations included the weak link between AR engagement and quiz outcomes, performance variability across topics, and the influence of external factors such as prior knowledge. Future upgrades could include enhanced instructional features, personalized learning paths, gamification, and collaborative learning elements. Conducting longitudinal studies and controlled trials would also help assess long-term effectiveness. In conclusion, while the AR application showed potential in fostering engagement, its impact on academic outcomes can be strengthened through better instructional design and interactivity.

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